

1. Hacking Web Servers

- Web servers host websites and web applications accessible via the internet.
 - They are common targets due to their public exposure.
 - Attackers exploit vulnerabilities in server software, configuration, or hosted applications to gain unauthorized access or disrupt services.
 - Common attacks:
 - Exploiting outdated software (unpatched OS or web server vulnerabilities).
 - Directory traversal attacks.
 - Remote code execution (RCE).
 - Misconfigured permissions.
 - SQL Injection affecting backend databases.
 - Cross-Site Scripting (XSS) that affects users but sometimes the server.
 - Securing web servers involves patching, proper configuration, minimal exposure of services, strong authentication, and logging.
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2. Web Application Vulnerabilities

- Web apps can be vulnerable due to programming flaws, poor input validation, or insecure design.
 - Common vulnerabilities include:
 - **SQL Injection (SQLi):** Malicious SQL commands executed by the database.
 - **Cross-Site Scripting (XSS):** Injecting malicious scripts into web pages viewed by others.
 - **Cross-Site Request Forgery (CSRF):** Tricks a user's browser into sending unwanted requests.
 - **File Inclusion Vulnerabilities:** Including unauthorized files (local or remote).
 - **Broken Authentication and Session Management:** Weak or exposed credentials.
 - **Security Misconfiguration:** Default passwords, verbose error messages.
 - OWASP Top 10 lists critical vulnerabilities and best practices.
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3. Web-Based Password Cracking Techniques

- Attackers target web authentication to crack passwords.

- Techniques include:
 - **Brute Force Attacks:** Trying all possible passwords.
 - **Dictionary Attacks:** Trying common password lists.
 - **Credential Stuffing:** Using leaked credentials from other breaches.
 - **Rainbow Tables:** Precomputed hash tables to reverse password hashes.
 - **Phishing:** Trick users into revealing passwords.
 - **Man-in-the-Middle (MITM):** Capture credentials on unencrypted connections.
 - Defenses include rate limiting, account lockouts, two-factor authentication, and strong password policies.
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4. Wireless Hacking

- Wireless networks use radio waves, introducing specific security challenges.
- Wireless hacking involves intercepting wireless communications or attacking wireless network protocols.
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Common attacks:

- **Eavesdropping:** Capturing unencrypted traffic.
 - **Man-in-the-Middle:** Intercept and alter communication.
 - **Rogue Access Points:** Fake APs trick users to connect.
 - **MAC Spoofing:** Changing MAC address to bypass filters.
 - **Denial of Service:** Jamming or flooding wireless signals.
 - **WEP/WPA Cracking:** Breaking weak encryption.
 - Wireless hacking tools: Aircrack-ng, Kismet, Wireshark.
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5. WEP, WPA Authentication Mechanisms and Cracking Techniques

- **WEP (Wired Equivalent Privacy):** Early wireless encryption, weak due to static keys and IV reuse; vulnerable to cracking by capturing enough packets.
- **WPA (Wi-Fi Protected Access):** Replaced WEP; uses TKIP and later AES (WPA2).
- **WPA Cracking:** Usually involves capturing the 4-way handshake during connection and using dictionary/brute force attacks.

- WPS (Wi-Fi Protected Setup) is also vulnerable to PIN attacks.
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6. Wireless Sniffers and Locating SSIDs, MAC Spoofing

- **Wireless sniffers** capture wireless packets.
 - **SSID:** Service Set Identifier, the network name broadcasted by AP.
 - Attackers use tools to detect SSIDs even if hidden.
 - **MAC Spoofing:** Changing the MAC address to impersonate a trusted device or bypass ACLs.
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7. Wireless Hacking Techniques

- **Packet sniffing and analysis.**
- **Deauthentication attacks:** Disconnecting users to capture handshake.
- **Rogue AP setup.**
- **WPS PIN brute forcing.**
- **Man-in-the-middle attacks using fake APs.**

- Jamming and signal interference.

8. Methods Used to Secure Wireless Networks

- Use **WPA2** or **WPA3** with strong AES encryption.
- Disable **WPS**.
- Use strong, complex passwords.
- Change default SSIDs and admin passwords.
- Use **MAC filtering** cautiously (not foolproof).
- Enable **network segmentation** and firewalls.
- Use **VPN** for sensitive communications.
- Regularly update firmware of routers.
- Monitor for rogue APs and suspicious activity.
- Use **802.1X authentication** with RADIUS servers for enterprise networks.